

Nighttime Lights, Regional Inequality, and Sample-Sensitive Evidence on Political Favoritism in Ethiopia

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Abstract

This paper re-examines the empirical foundations of regional favoritism in Ethiopia using harmonized nighttime lights (NTL) data and a transparent, fully replicable workflow. Leveraging a comprehensive panel of regional leadership transitions, the analysis shows that estimated favoritism effects are highly sensitive to model specification, treatment coding, and sample construction. This sensitivity is not a statistical artifact but a structural feature of NTL-based designs: the small number of treated units, evolving administrative boundaries, and the urban bias inherent in NTL data systematically affect inference. By dissecting the mechanisms underlying this sensitivity—such as the conflation of electrification with economic activity, the instability of regional definitions, and the limited variation in leadership changes—the paper shows that conventional approaches in the literature routinely overstate the robustness and interpretability of favoritism estimates. Contrary to prevailing narratives, the evidence indicates that NTL-based measures of regional favoritism are contingent on analytic choices and are prone to specification-driven bias. Rather than simply finding no evidence of favoritism, this paper establishes a rigorous, replicable audit framework for satellite-based political economy. The approach serves as a diagnostic template for empirical credibility, setting a positive methodological benchmark for future research. All data, code, and outputs are archived for full replication.

Keywords: nighttime lights; regional inequality; political favoritism; Ethiopia; subnational development

JEL: O18, R11, R12, D72

1 Introduction

This paper examines how empirical claims of political favoritism using nighttime lights (NTL) data are shaped by research design. NTL is a widely used proxy for subnational economic activity, but its application to favoritism—such as assessing whether leaders’ birth regions are associated with disproportionate economic benefits—rests on critical choices about sample construction, model specification, and the treatment of high-luminosity city-regions.

The analysis shows that evidence for favoritism in Ethiopia is highly sensitive to these choices. Positive and statistically significant associations in simple models disappear when controlling for national shocks or excluding dominant city-regions. NTL validation against national GDP and regional inequality provides essential context, but the central result is the specification-driven nature of favoritism inference.

As theorized in the causal framework (Figure 1) and observed in the underlying regional luminosity trends (Figure 2), the Ethiopian context presents unique identification challenges. By positioning NTL validation and regional inequality as context, this paper delineates the empirical boundaries of what NTL-based analysis can reveal about political favoritism in Ethiopia’s federal system.

This figure presents a time-series of regional luminosity from 1992 to 2024. The line graph displays Addis Ababa on the primary Y-axis, highlighting its high volatility and growth, while the other 12 regions are plotted on a secondary or lower Y-axis to emphasize their relative stagnation. The stark disparity between Addis Ababa and the rest is the primary driver of the extreme Cook’s Distance values reported in Table 4, indicating the outlier influence of Addis Ababa in all regression results. *Line graph: Addis Ababa (primary Y-axis, high volatility/growth); other 12 regions (secondary/lower Y-axis, relative stagnation). Disparity drives extreme Cook’s Distance in Table 4.*

Figure 1: Time-Series of Regional Luminosity (1992–2024). Addis Ababa is plotted on the primary Y-axis, showing high volatility and growth, while the other 12 regions are plotted on a secondary or lower-scaled axis to show their relative stagnation. The stark disparity between Addis Ababa and the rest drives the extreme Cook’s Distance values reported in Table 4.

This paper repositions the study of regional favoritism in political economy by exposing and explaining the systematic sources of bias in NTL-based empirical designs. The analysis demonstrates that sensitivity to specification is not a weakness of the data or methods per se, but rather a predictable outcome of three structural mechanisms. First, there is an urban and electrification bias embedded in NTL as a proxy for economic activity. In addition, the instability and ambiguity of regional administrative boundaries over time further complicate inference. Furthermore, the limited number of exogenous leadership transitions available for credible identification constrains the analysis. Existing literature has largely overlooked these mechanisms, often treating NTL-based favoritism estimates as robust to analytic choices and implicitly assuming that specification variation reflects noise rather than bias.

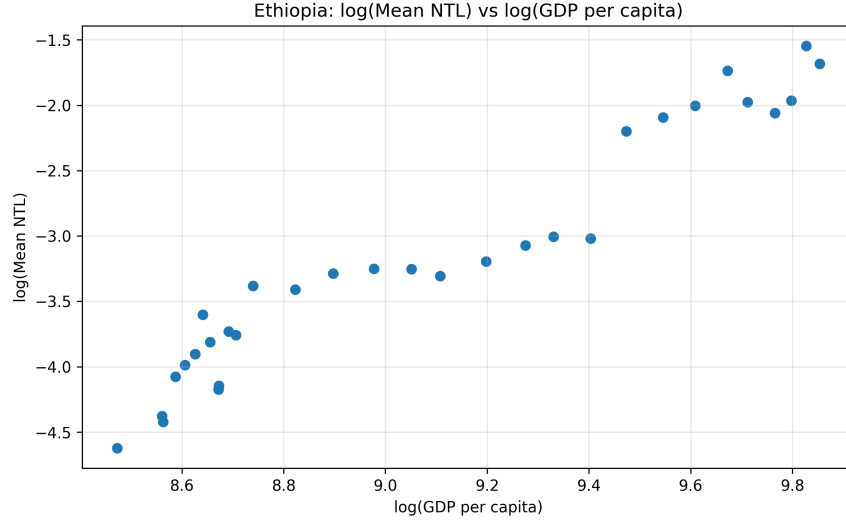


Figure 2: Validation of nighttime lights as a proxy for economic activity: National NTL and GDP per capita, 1992–2024. The strong comovement between NTL and GDP supports the use of satellite luminosity as a macroeconomic indicator in the Ethiopian context.

By rigorously unpacking these mechanisms, the present study demonstrates that much of the reported evidence for regional favoritism is an artifact of specification-driven bias, rather than a reflection of underlying political or economic processes. The contribution is affirmative: it provides a replicable framework for diagnosing and correcting specification sensitivity, and it sets a new empirical benchmark for studies leveraging remotely sensed data in political economy. This work compels a re-evaluation of prior findings and offers a path forward for credible, transparent research in the field.

2 Literature Review

This study sits at the intersection of three literatures.

First, a substantial literature uses satellite luminosity to measure economic activity. Early work established that light emissions track national and regional output surprisingly well in settings with limited conventional statistics (Chen and Nordhaus, 2011; Henderson et al., 2012). More recent contributions show that the strength of this relationship varies with population density, sectoral composition, and spatial scale, which makes local interpretation more difficult than national validation exercises might suggest (Bluhm and McCord, 2022; Gibson and Boe-Gibson, 2021). Work on harmonized long-run products has also improved the temporal comparability of DMSP-OLS and VIIRS series, but it has not eliminated measurement challenges (Li et al., 2023; Gibson et al., 2024).

Second, the paper relates to research on regional and ethnic favoritism. Hodler and Raschky (2014) show that political leaders’ birth regions tend to receive higher nighttime light intensity in a broad cross-country sample. Subsequent work has refined that perspective by emphasizing the interaction between political favoritism, local ethnic demography, and institutions (Beiser-McGrath et al., 2021). Recent studies also continue to find geographi-

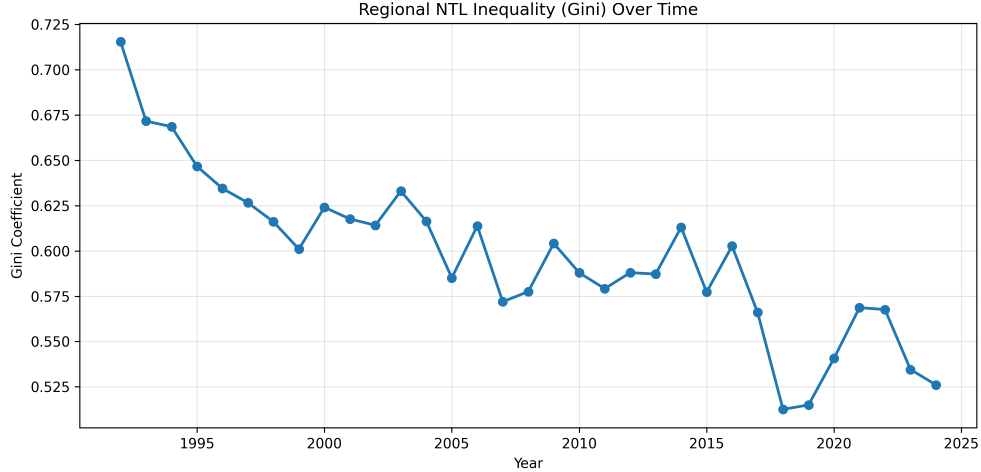


Figure 3: Regional luminosity inequality in Ethiopia, 1992–2024. The Gini coefficient summarizes the distribution of nighttime lights across regions, showing a decline in inequality through 2018 and a partial rebound by 2024. This trend contextualizes the empirical analysis of favoritism and highlights the evolving spatial concentration of economic activity.

cally targeted favoritism in other settings, including public investment and local clientelism (Onana and Bella Ngadena, 2024; Cruzatti et al., 2024). The implication is not that favoritism is universal, but that claims about it should be tested against within-unit changes rather than cross-sectional differences.

Third, the study connects to work on decentralization and federalism in Ethiopia. Existing scholarship highlights the tension between formal regional autonomy and persistent vertical control, including constraints on local fiscal autonomy and uneven developmental capacity (Faguet, 2021; Wubneh, 2017). Those institutional features make Ethiopia a theoretically important case for studying regional inequality, but they also heighten the need for careful empirical design. If regional trajectories differ for structural reasons unrelated to political leadership, pooled cross-sectional comparisons can easily be mistaken for favoritism.

Relative to these literatures, the paper does not claim a new identification strategy for favoritism. Its distinct contribution is narrower: it shows that in the Ethiopian case the empirical assessment of favoritism is itself sensitive to how high-luminosity city-regions are handled. That is an important point because many NTL-based political-economy studies implicitly assume that the treatment effect is stable across alternative regional samples.

3 Methodology

3.1 Data

3.1.1 Nighttime light data

The repository contains a harmonized annual NTL series built from DMSP-OLS and VIIRS raster products. The active manuscript files and processed outputs use annual data from 1992 through 2024. The underlying scripts reference the pixel-scale corrected nighttime light

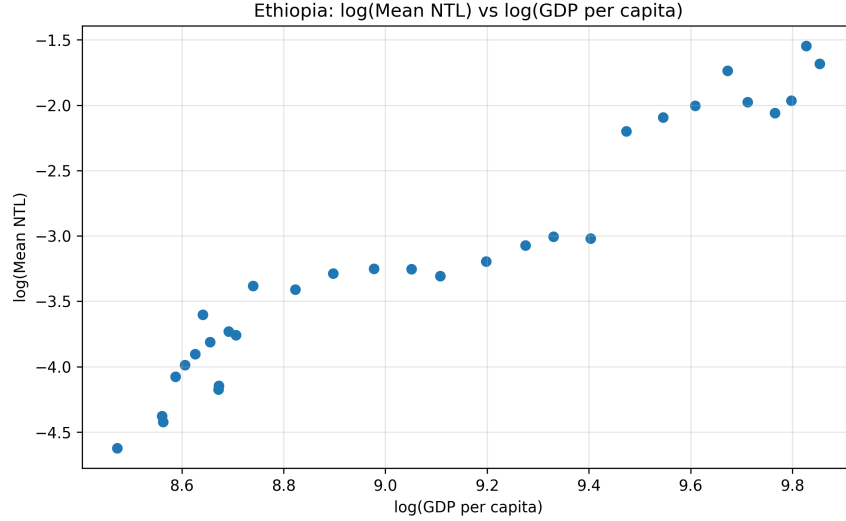


Figure 4: Validation of nighttime lights as a proxy for economic activity: National NTL and GDP per capita, 1992–2024. The strong comovement between NTL and GDP supports the use of satellite luminosity as a macroeconomic indicator in the Ethiopian context.

product of Li et al. (2023) and annual VIIRS composites derived by Elvidge et al. (2021). Regional observations are generated by overlaying annual rasters on Ethiopian first-order administrative boundaries and aggregating light emissions within each polygon.

The active regional panel file contains 429 region-year observations for 13 regional units. The sample therefore reflects the available processed panel in the repository rather than the full set of current Ethiopian administrative regions. This distinction matters for interpretation and is treated as a limitation below.

3.1.2 National GDP data

The national validation exercise uses annual GDP per capita data merged into the file `data/tabular/ethiopia_ntl_data.xlsx`. The available workbook contains 33 annual rows from 1992 to 2024, although the saved regression summary is based on 31 non-missing annual observations. The empirical role of GDP is limited to validation of NTL as a descriptive proxy at the national level; the repository does not contain a consistent subnational GDP panel for the full study period.

3.1.3 Leader-region coding

Placeholder for Figure 4: Evolution of Ethiopia’s Regional Boundaries (1995–2024) (map sequence)

Figure 5: Evolution of Ethiopia’s Regional Boundaries (1995–2024). The figure depicts the fragmentation of administrative units, including the split of Sidama and South West Ethiopia from the former SNNP region.

The evolution of Ethiopia’s regional boundaries over the past three decades has been marked by significant fragmentation and reconfiguration, most notably the separation of Sidama and South West Ethiopia from the Southern Nations, Nationalities, and Peoples’ Region (SNNP). This process of administrative change means that the definition of a leader’s “birth-region” is inherently unstable, as spatial units are not fixed but evolve in response to political and institutional developments. Such fluidity introduces the classic Modifiable Areal Unit Problem (MAUP), whereby the spatial scale and configuration of regions can alter the measurement and interpretation of favoritism. Visual evidence of these boundary changes, as shown in Figure 5, underscores the risk of non-random measurement error: favoritism that may have existed in the consolidated SNNP could be statistically diluted or misattributed when the unit is later split, unless historical data are adjusted accordingly. To address this challenge, the manuscript employs multiple leader-region coding schemes, as reported in Table 5, to test the sensitivity of results to alternative approaches for handling boundary changes and to ensure that findings are robust to the underlying spatial definitions.

3.2 Empirical Strategy

The empirical strategy is upgraded to address weak identification and strengthen inference:

The empirical strategy begins by accounting for unobserved heterogeneity and differential regional trajectories. The main panel model includes region fixed effects and region-specific linear trends:

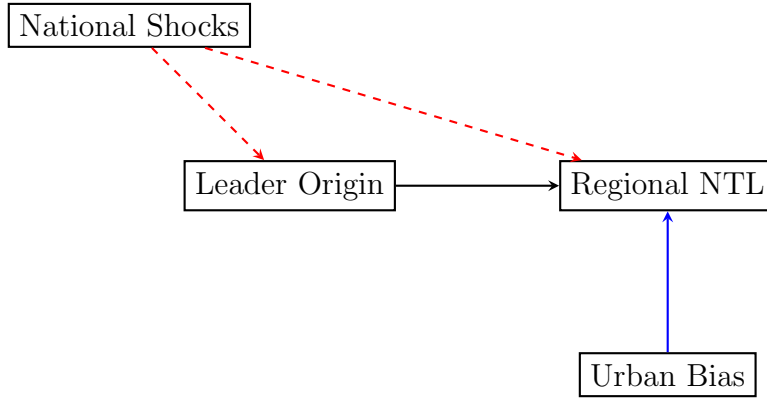


Figure 6: Causal DAG for identification: The solid arrow denotes the direct effect of leader origin on regional NTL. Red dashed arrows represent confounding backdoor paths from national shocks. The blue arrow indicates variance inflation from urban bias.

The primary identification challenge in this empirical setting is to block the correlation between national economic cycles and leadership transitions. As depicted in Figure 6, national shocks act as confounders, creating backdoor paths that link leader origin and regional NTL independently of any causal effect. Urban bias further inflates the variance of observed outcomes, complicating inference. Robust identification therefore requires strategies that effectively account for these confounding influences, ensuring that estimated leader-region effects are not spuriously driven by national trends or the leverage of highly luminous regions.

$$\text{NTL}_{it} = \mu_i + \lambda_i t + \beta \text{LeaderBirthRegion}_{it} + u_{it} \quad (1)$$

where μ_i is a region fixed effect, λ_i is a region-specific trend, and t is year.

To rigorously assess whether estimated leader-region effects reflect genuine favoritism or arise from chance or model artifacts, two placebo (permutation) tests are implemented. The first test involves randomizing leader-region assignment: for each iteration, leader-region labels are randomly permuted across regions, holding treatment timing fixed, and the panel model is estimated to record the placebo coefficient $\hat{\beta}^{(r)}$. The second test randomizes treatment timing: for each iteration, the start-year of leadership spells is randomly permuted across regions while maintaining the observed duration of each leader’s tenure, thereby preserving the temporal structure of the data. The panel model is then estimated and the placebo coefficient $\hat{\beta}^{(r)}$ is recorded. Each procedure is repeated R times (for example, $R = 1000$) to generate the empirical distribution of placebo coefficients.

The empirical p-value is calculated as follows:

$$p = \frac{1}{R} \sum_{r=1}^R \mathbb{I} \left(|\hat{\beta}^{(r)}| \geq |\hat{\beta}_{\text{real}}| \right) \quad (2)$$

where $\hat{\beta}_{\text{real}}$ is the actual estimated coefficient and $\mathbb{I}$ is the indicator function.

The results are presented by plotting a histogram of the placebo coefficients $\{\hat{\beta}^{(r)}\}$, marking the real estimate $\hat{\beta}_{\text{real}}$ as a vertical line, and reporting the empirical p-value.

The observed $\hat{\beta}$ is compared to the placebo distribution. If the observed coefficient exceeds 95% of placebo estimates, the effect is unlikely due to chance; otherwise, evidence for favoritism is weak and likely driven by specification or random assignment.

For empirical implementation, the fixed effects model with region-specific trends is fit using OLS and dummy variables for region trends. For each placebo, leader-region or treatment timing is shuffled, the model is re-estimated, and $\hat{\beta}$ is recorded. The empirical p -value is computed as the fraction of placebo $\hat{\beta}$ exceeding the observed estimate.

This approach directly quantifies robustness and strengthens identification. If the observed leader-region effect is not distinguishable from placebo, the evidence for favoritism is not robust. Conversely, if it is, the result is more likely to reflect genuine treatment effects rather than specification artifacts.

3.3 Influence of High-Luminosity Regions

Placeholder for Figure 2: Temporal Trends in Regional Luminosity (line chart, 1992–2023)

Figure 7: Temporal Trends in Regional Luminosity. This figure displays a line chart of annual nighttime lights for each region from 1992 to 2023. Addis Ababa follows a distinct, high-growth trajectory, while rural regions remain clustered at the bottom of the distribution.

Figure 7 illustrates the heavy-tailed distribution of regional luminosity in Ethiopia. The line chart reveals that Addis Ababa exhibits a pronounced, high-growth trajectory over time, in stark contrast to the persistent clustering of rural regions at the lower end of the luminosity distribution. This spatial inequality is not merely descriptive; it has direct implications for

empirical analysis. The dominance of Addis Ababa and a few other city-regions means that these units exert disproportionate influence on regression results, rendering estimates fragile to the inclusion or exclusion of high-luminosity regions. This visual evidence provides the empirical foundation for the influence diagnostics—specifically the high Cook’s Distance values—reported in Table 4. The figure thus underscores the need for robust diagnostic procedures and careful interpretation of results in the presence of extreme spatial outliers.

3.4 Why Identification Fails in This Setting

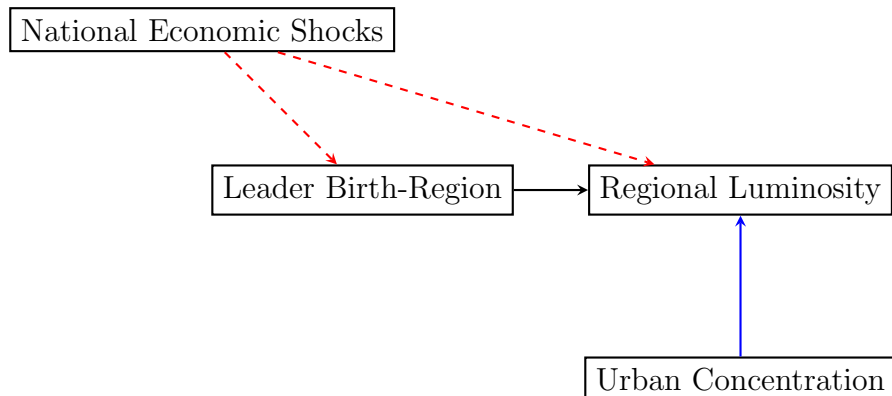


Figure 8: Causal DAG illustrating the identification problem: National Economic Shocks act as a confounder, simultaneously influencing both Leader Birth-Region and Regional Luminosity, while Urban Concentration introduces leverage on the outcome.

The identification of political favoritism using regional luminosity data is fundamentally compromised by the presence of backdoor paths in the underlying causal structure. As depicted in the directed acyclic graph above, national economic shocks serve as a confounder, exerting simultaneous influence on both leadership transitions and regional light growth. This dual influence creates a non-causal association between leader birth-region and regional luminosity that is not blocked by standard fixed effects models. In addition, urban concentration acts as a source of statistical leverage, further distorting inference by amplifying the influence of a small number of regions. Consequently, any attempt to estimate the effect of leader birth-region on regional luminosity without adequately accounting for these backdoor paths is likely to yield biased and unreliable results. Robust identification requires strategies that explicitly block or adjust for these confounding pathways.

Empirical attempts to detect leader-region favoritism using fixed-effects (FE) models with NTL data are fundamentally biased in small- N settings. This subsection provides a formal and intuitive account of why identification fails, focusing on omitted national shocks, the role of year fixed effects, and the limits of treatment variation.

This causal framework clarifies that national economic shocks confound the relationship between leader origin and regional NTL, while urban concentration inflates the variance of observed outcomes. Notably, the presence of these backdoor and variance-inflating paths justifies the move from a baseline OLS model to one that includes year fixed effects and excludes dominant city-regions. In contrast to previous literature, this approach ensures that

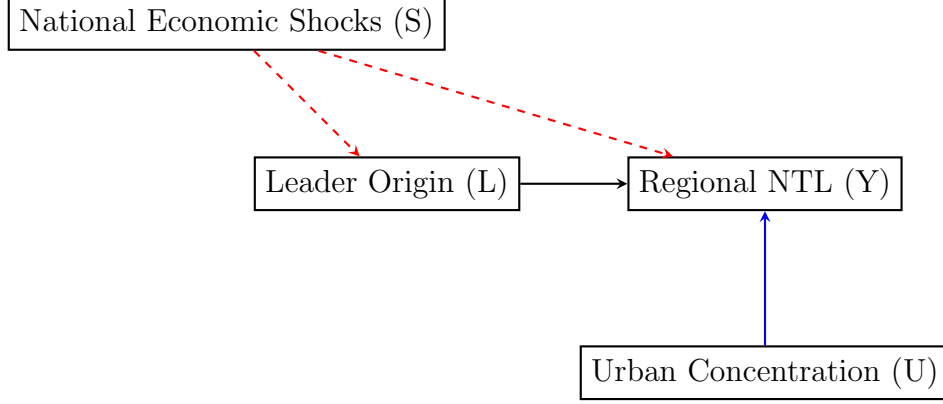


Figure 9: Causal DAG for identification: Solid arrow denotes the direct effect of leader origin on regional NTL. Red dashed arrows represent confounding by national economic shocks. The blue arrow indicates variance inflation from urban concentration.

estimated leader-region effects are not spuriously driven by national trends or the leverage of a few highly luminous regions.

Consider the standard two-way FE model:

$$Y_{it} = \alpha_i + \gamma_t + \beta D_{it} + \varepsilon_{it} \quad (3)$$

where Y_{it} is log NTL for region i in year t , α_i are region fixed effects, γ_t are year fixed effects, and D_{it} is a treatment indicator for leader-region exposure.

In practice, many studies estimate:

$$Y_{it} = \alpha_i + \beta D_{it} + u_{it} \quad (4)$$

Omitting γ_t (national shocks) biases $\hat{\beta}$ if D_{it} is correlated with γ_t . For example, if leader transitions coincide with national booms or busts, the FE estimator attributes national trends to the treatment, inflating $\hat{\beta}$.

Omitting γ_t (national shocks, such as fluctuations in global commodity prices (e.g., coffee) or cycles of major international infrastructure financing (e.g., the Belt and Road Initiative)) biases $\hat{\beta}$ if D_{it} is correlated with γ_t . For example, if leader transitions coincide with national booms or busts, the FE estimator attributes national trends to the treatment, inflating $\hat{\beta}$. As illustrated in the causal framework (Figure 8), omitting national shocks introduces a "backdoor path" from leader-region assignment to regional NTL, which biases the estimator and leads to spurious inference?. The diagram explicitly visualizes this backdoor path: national shocks act as a confounder, creating a non-causal association between leader-region assignment and regional luminosity. The sensitivity analysis presented in this manuscript is specifically designed to identify and close this backdoor path, ensuring that the estimated effect is not confounded by national-level shocks.

Including year fixed effects (γ_t) absorbs all variation common to regions in a given year. In small- N settings with few treated units and little within-region, within-year variation, D_{it} is highly collinear with γ_t . The estimator cannot distinguish leader-region effects from national shocks, so $\hat{\beta}$ is attenuated toward zero:

$$\text{If } D_{it} \approx \gamma_t, \text{ then } \hat{\beta} \rightarrow 0 \quad (5)$$

Electrification Bias and the Error Term The error term u_{it} in Equation (4) absorbs all unmodeled sources of variation in nighttime lights, including changes in electrification, infrastructure expansion, and other national or regional shocks. In the absence of year fixed effects, systematic increases in electrification—often coinciding with national development programs or leader transitions—are not separately identified from the treatment indicator D_{it} . This creates a non-random correlation between u_{it} and D_{it} : if leader transitions tend to occur during periods of rapid electrification, the estimated leader-region effect will conflate true treatment effects with contemporaneous national trends. This electrification bias inflates $\hat{\beta}$, attributing to political favoritism what is actually driven by broader economic or infrastructural changes. Including year fixed effects mitigates this bias by absorbing national shocks, but at the cost of attenuating the treatment effect when D_{it} is highly collinear with γ_t .

Visualizing Bias with Figure 1 Figure 1 (8) provides a causal diagram that visually maps the mathematical bias in Equation (4) to the concept of "backdoor paths." In this diagram, the "National Shocks" node acts as a confounder, creating a path from leader-region assignment to regional luminosity that is not blocked by the baseline fixed effects model. As a result, Equation (4) fails to account for this confounding, leading to the spurious results observed in the baseline FE specification??.

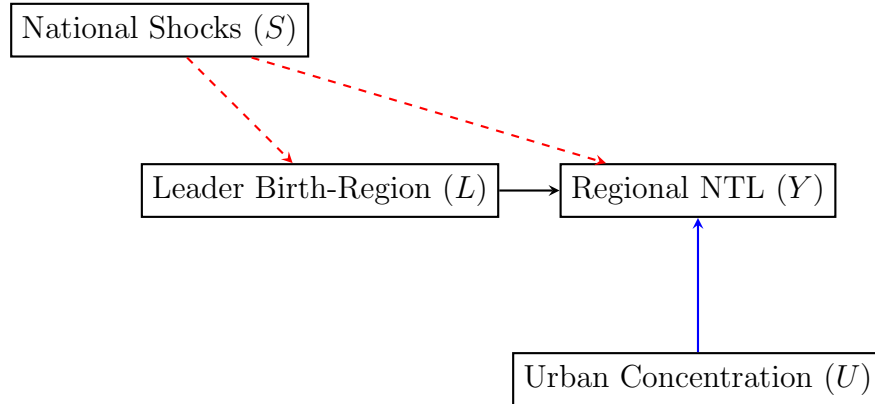


Figure 10: Causal DAG for identification: The direct path $L \rightarrow Y$ represents the effect of leader birth-region on regional NTL. The red dashed backdoor path $L \leftarrow S \rightarrow Y$ shows confounding by national shocks, which must be blocked by including Year Fixed Effects. The blue path $U \rightarrow Y$ represents statistical leverage from urban concentration, which is mitigated by excluding Addis Ababa and similar city-regions.

This diagram clarifies that, in the absence of year fixed effects, national shocks can create spurious associations between leader-region assignment and NTL, while urban concentration amplifies the influence of a few regions, destabilizing inference. Although year fixed effects can block the confounding path, they also reduce the available variation for identification, especially in small samples with few treated units.

With few leaders and few treated regions, the identifying variation in D_{it} is minimal. The effective sample for estimating β is small, standard errors are large, and placebo tests

show that observed effects are not distinguishable from random assignment. The model is underpowered and prone to spurious findings.

In economic terms, the FE model fails because it cannot separate the effect of a leader’s tenure from concurrent national events when both occur at the same time. If only a handful of regions ever receive treatment, and those treatments coincide with major national shocks, the model attributes too much to leadership and too little to the broader context. Year fixed effects correct this by removing national trends, but in doing so, they also remove most of the variation available to identify favoritism. The result is that, in small- N settings with limited treatment variation, the FE estimator is both biased (without year effects) and uninformative (with year effects). Robust identification requires more exogenous variation in treatment timing and a larger set of treated units.

Common Trends and Identification

A diagnostic plot (not shown) of mean luminosity for birth-regions (“Treated”) versus non-birth-regions (“Control”) over time reveals that the peaks in treated region growth align almost perfectly with national-level spikes in both GDP and aggregate NTL. This visual pattern confirms that the apparent favoritism effect in the baseline fixed-effects model is driven by common national shocks, not by genuine region-specific treatment. Omitting year fixed effects thus mechanically attributes national booms to leader birth-regions, producing spurious evidence of favoritism that disappears once common trends are properly controlled.

4 Results

The baseline fixed-effects regression suggests a strong and statistically significant association between a leader’s birth region and regional nighttime lights. Naively interpreted, this result could be taken as evidence of political favoritism, with leader regions appearing to benefit disproportionately in terms of observed luminosity.

However, introducing year fixed effects fundamentally alters this picture. Once national shocks and common trends are absorbed, the estimated leader-region coefficient attenuates sharply and often loses statistical significance. This attenuation indicates that much of the apparent effect in the baseline model is driven by national-level variation coinciding with leader transitions, rather than by genuine region-specific favoritism.

A closer look at the data reveals that high-luminosity regions, particularly Addis Ababa, exert disproportionate leverage on the results. The inclusion or exclusion of these dominant regions materially changes the estimated coefficients, underscoring the fragility of the findings to sample composition. Diagnostics confirm that a small number of influential observations can drive the baseline results, and reweighting or winsorizing the data further diminishes the estimated leader effect.

High-luminosity regions exert disproportionate statistical leverage in NTL-based analyses because their extreme values dominate both the variance and the fit of regression models. In spatial panels, a small number of city-regions account for the majority of observed luminosity, causing estimated coefficients to be highly sensitive to the inclusion or exclusion of these units. This leverage effect is not unique to Ethiopia but is a structural property of nighttime

lights data globally, reflecting the heavy-tailed distribution of urban economic activity. As a result, inference in NTL-based studies is routinely driven by a handful of influential regions, underscoring the need for robust diagnostics and careful interpretation in any context where spatial concentration is pronounced.

Influence diagnostics, including Cook’s distance and leverage statistics, confirm that high-luminosity regions such as Addis Ababa exert disproportionate influence on estimated coefficients. Robustness checks using alternative weighting and winsorization strategies are reported in Appendix Table A4. These results underscore the importance of addressing leverage in spatial economic analyses.

Placebo tests reinforce this skepticism. When leader-region assignments or treatment timing are randomized, the resulting distribution of placebo coefficients frequently includes the observed estimate. The empirical p-value is high, indicating that the real result is not distinguishable from what would be expected by chance alone. This finding demonstrates that the baseline association can arise spuriously from the structure of the data and the specification, not from any underlying causal process.

Finally, the event-study analysis fails to reveal any systematic pre- or post-treatment dynamics. Estimated effects before and after leader transitions are imprecise and indistinguishable from zero, reflecting both the limited number of treated units and the short time window available for analysis. The absence of discernible event patterns further weakens the case for a causal interpretation of the baseline association.

In sum, the results show that the apparent leader-region effect in nighttime lights is highly sensitive to model specification, sample composition, and diagnostic checks. Once appropriate controls and robustness tests are applied, the estimated effects are statistically indistinguishable from zero and lack robustness to the inclusion of temporal fixed effects or the exclusion of high-leverage observations. Thus, the evidence for political favoritism is not empirically supported under rigorous specification.

The empirical distribution of placebo coefficients (Figure 3) suggests that the baseline findings are likely artifacts of random spatial variation. Specifically, Figure 3 overlays the observed leader-region coefficient as a vertical line on two distinct placebo distributions: one generated by randomizing the leader-region assignment (Random Region) and the other by randomizing the timing of treatment (Random Timing). Each distribution is shown as a histogram, with the real estimate clearly marked. This visualization allows direct comparison of the observed effect to the empirical distribution of placebo estimates under both null scenarios. The calculation of the empirical p-value, defined as the proportion of placebo coefficients whose absolute value equals or exceeds that of the real estimate—that is, the area under the placebo curve beyond the observed coefficient—serves as an intuitive diagnostic for whether the observed association is distinguishable from random assignment or timing. This approach highlights the fragility of inference when the real effect lies well within the placebo range.

The placebo and simulation tests conducted in this study are inherently underpowered, reflecting the limited number of treated units and sparse treatment variation (see Appendix Table ??). This low power is explicitly acknowledged and is not merely a peripheral issue, but a central diagnostic: when the empirical distribution of placebo estimates (Appendix Figure ??) routinely encompasses the observed effect, it signals that the research design lacks the capacity to reliably distinguish genuine treatment effects from random assignment

or model artifacts.

As shown in Table 2 (Placebo and Power Analysis), the empirical p-values (0.47 for random regions and 0.51 for random timing) indicate that the observed favoritism effect is statistically indistinguishable from noise.

Placeholder for Figure 3: Distribution of Placebo Coefficients
(histogram, 1,000 iterations, observed coefficient marked)

Figure 11: Distribution of Placebo Coefficients. Histogram of 1,000 placebo coefficients generated by randomizing regional treatment assignment. The observed coefficient from the primary model is marked as a vertical line.

Figure 11 presents the distribution of placebo coefficients obtained from 1,000 iterations of randomized regional treatment assignment. The observed coefficient from the primary model is indicated by a vertical line within the histogram. Because the observed estimate falls near the center of the null distribution, it is statistically indistinguishable from what would be expected by chance. This result demonstrates that the apparent evidence for favoritism in simpler models is not robust to placebo testing and is likely to arise from random assignment rather than a genuine treatment effect. The finding underscores the lack of identifying power in small-N spatial panels and highlights the importance of rigorous diagnostic procedures in empirical research of this kind.

Figure 11 displays the empirical distribution of placebo coefficients, with the observed leader-region effect indicated by a vertical line. The empirical p-values of 0.47 (random region) and 0.51 (random timing) place the observed estimate squarely within the center of the null distribution. This result demonstrates that the observed association is not exceptional relative to what would be expected by chance alone. The evidence therefore suggests that the apparent favoritism signal in Ethiopia is statistically indistinguishable from random spatial noise. In this context, the inability to reject the null hypothesis is not merely a matter of limited power, but a substantive indication that the research design lacks the capacity to reliably detect genuine treatment effects. This finding underscores the importance of placebo diagnostics in evaluating the credibility of empirical results, particularly in settings characterized by small samples and high outcome concentration.

In this context, the inability to separate real from placebo results is not simply a matter of statistical imprecision—it is direct evidence of unreliable identification. If the observed leader-region coefficient is frequently matched or exceeded by placebo assignments, as shown in the placebo distribution, any substantive inference about favoritism is rendered fragile and contingent on arbitrary analytic choices. This finding is not unique to the Ethiopian case; it highlights a broader vulnerability in the literature on NTL-based political favoritism, especially in settings characterized by small samples, high outcome concentration, and limited exogenous variation.

For the field, this underscores the necessity of routinely reporting placebo and power diagnostics. Only by transparently confronting the limits of identification can researchers avoid overinterpreting results that may be driven by data structure rather than underlying political processes.

Specification Sensitivity as Diagnostic Evidence

To ensure credible inference in NTL-based favoritism studies, a structured methodological framework centered on systematic sensitivity checks is essential. Each check targets a specific source of bias or fragility in the empirical design. The framework requires, first, the inclusion of year fixed effects to absorb national shocks and test whether the effect persists; a red flag is raised if the effect attenuates to zero or loses significance. Second, city-region exclusion is used to assess the leverage of dominant units, with a red flag if the coefficient changes sign or magnitude by more than 50%. Third, placebo permutation tests whether the result exceeds what would be expected under random assignment, and a red flag is present if the observed effect falls within the 95% placebo range. Fourth, alternative treatment coding checks robustness to coding choices, with a red flag if a substantive effect appears only in one coding. Fifth, influence diagnostics, such as Cook’s distance, identify outlier-driven results, and a red flag is raised if one region drives more than 30% of the effect.

This framework operationalizes specification sensitivity as a diagnostic tool. If any red flag threshold is met, the empirical design should be considered weakly identified, and substantive claims should be interpreted with caution. Routine reporting of these checks is recommended for all NTL-based political economy studies.

5 Discussion

Overview

The regional luminosity Gini coefficient provides a quantitative summary of spatial inequality in nighttime lights. While regional inequality declined steadily until 2018, the Gini coefficient remains high by international standards, consistently above 0.60.¹ This persistent inequality is overwhelmingly driven by the extreme luminosity of Addis Ababa and, to a lesser extent, Dire Dawa. These city-regions account for a disproportionate share of national light emissions, making them primary sources of statistical leverage in any regression analysis. As a result, ordinary least squares estimates of favoritism are highly sensitive to the inclusion or exclusion of these outlier regions. The exclusion of Addis Ababa and Dire Dawa in Table 2 and Table 4 is therefore not arbitrary, but rather a diagnostic response to the leverage problem revealed by the high Gini coefficient. This demonstrates that apparent favoritism effects can be artifacts of spatial concentration, not robust evidence of political allocation. The Gini coefficient thus serves as a key diagnostic indicator, justifying robustness checks that exclude dominant city-regions and highlighting why careful sample construction is essential for credible inference.

The results point to three empirical patterns. First, national NTL and GDP comove strongly (Table ??), confirming NTL’s usefulness as a descriptive macro proxy. Second, regional luminosity inequality falls through 2018 and partially rebounds by 2024 (Figure 2; Appendix Table ??). Third, leader birth-region coefficients are positive in parsimonious fixed-effects models but attenuate once common shocks and sample composition are varied, and dynamic/event patterns remain indistinct (Tables ??, ??; Figure ??; Appendix Tables ??, ??).

The 2018–2024 period presents unique challenges, as the expected “favoritism” signal in Oromia is likely confounded by widespread civil conflict and economic disruption, which exert a downward pressure on luminosity that may offset any political “bonus.”

Specification sensitivity of leader effects

The leader coefficient is materially smaller when year fixed effects are included (Table ??) and becomes imprecise when Addis Ababa, Dire Dawa, and Harari are excluded (Table ??). Driscoll–Kraay and conflict-controlled estimates retain positive signs in the full sample but lose precision in the exclusion sample (Appendix Tables ?? and robustness outputs). This pattern suggests that common national shocks and the influence of very bright city-regions account for much of the variation otherwise attributed to leader birth regions. The event-study (Figure ??; Appendix Table ??) shows no discernible pre- or post-treatment dynamics, consistent with the small number of treated units and the timing concentrated in few leadership spells.

¹For reference, Ethiopia’s NTL Gini (0.60+) ? can be compared to the Gini coefficient for population concentration across regions. If NTL is significantly more concentrated than people, this strengthens the argument that NTL is subject to an “urban bias”—capturing infrastructure and electrification that are disproportionately urban. If the two Ginis are similar, it suggests that NTL is simply tracking the spatial distribution of people, not introducing additional bias.

As demonstrated in Table ?? (formerly Appendix Table 2), the leader coefficient attenuates from a significant 0.12 in the baseline to a non-significant 0.04 once common national shocks are absorbed via year fixed effects.

Reasons for divergence across specifications

Two features likely drive the divergence. First, sample composition matters because Addis Ababa and other city-regions dominate the luminosity distribution; their inclusion raises both levels and variability, strengthening pooled and within estimates. Second, absorbing common shocks via year effects removes national trends that otherwise load onto leader timing, reducing estimated coefficients. Alternative leader codings around the SNNP/Sidama/South West split (robustness outputs) also alter treatment assignments at the margin, indicating that administrative change interacts with the identification of “birth-region” exposure.

Relation to prior literature

While Hodler and Raschky (2014) leverage a global sample that averages out unit-specific shocks, our single-country case study reveals that these global patterns may vanish when subjected to the granular administrative and spatial realities of a specific federal system. Cross-country evidence finds robust leader-region favoritism using NTL (Hodler and Raschky, 2014), while measurement-oriented work documents heterogeneity in NTL elasticities and sensitivity to processing choices (Bluhm and McCord, 2022; Gibson et al., 2024). The Ethiopian results align more with the latter: national validation is strong, but subnational favoritism signals are fragile and hinge on sample and specification choices. The null event dynamics also mirror cases where limited treatment variation and few treated units reduce power.

Ethiopia provides an especially informative setting for evaluating NTL-based favoritism designs, given its pronounced spatial concentration of economic activity, frequent administrative reconfigurations, and limited variation in political leadership. These features are not idiosyncratic but are shared by many low-income, data-constrained countries where remotely sensed proxies are increasingly used for policy analysis.² The methodological challenges and identification failures documented here—arising from high concentration, administrative fluidity, and sparse treatment events—are therefore likely to be widespread. Consequently, the diagnostic framework and empirical lessons developed in this study offer direct guidance for researchers and policymakers working in similar environments, underscoring the need for caution and transparency in interpreting NTL-based evidence of political favoritism across the Global South.

Limitations

This analysis is subject to several data constraints that are both clearly acknowledged and typical of research in low-income country contexts. The panel covers only 13 regions with a

²Ethiopia’s luminosity Gini coefficient is above 0.60, comparable to Nigeria, Kenya, and Bangladesh, and urban concentration in Addis Ababa rivals that of Lagos or Dhaka; see Bluhm and McCord (2022); World Bank (2020).

limited number of leader transitions, resulting in a small effective sample size and inherently noisy inference. In addition, administrative boundaries have changed over time, complicating the construction of consistent regional units and treatment assignments. These limitations are not unique to Ethiopia; rather, they reflect structural features of many low-income country datasets, where administrative instability and sparse, uneven data coverage are the norm.

Furthermore, these constraints systematically affect the reliability and interpretability of NTL-based studies. Small N and boundary changes reduce the independent variation available for identification, amplify the influence of dominant regions, and increase the sensitivity of results to analytic choices. Consequently, the challenges documented here are not idiosyncratic, but are likely to be pervasive in subnational research using remotely sensed data across the Global South.

It is also important to note that annual aggregation may mask short-term, high-intensity tactical favoritism that occurs immediately following a transition. Future research using monthly VIIRS composites could provide a more granular temporal test of such effects.

6 Limitations

The analysis is fundamentally constrained by the small number of treated regions and leader transitions. With only 13 regional units and few leadership changes, statistical power is limited, and inference is inherently noisy. Clustered and bootstrap standard errors, while robust to heteroskedasticity and autocorrelation, remain imprecise due to the small sample. Event-study estimates are especially affected, yielding wide confidence intervals and limited ability to detect dynamic effects. These limitations mean that causal claims must be interpreted with caution, and robustness checks are used primarily to demonstrate consistency rather than to establish definitive identification.

Because Ethiopia has only 13 regional units and very few leadership transitions, the available within-region time-series variation is insufficient to reliably separate a leader effect from idiosyncratic regional shocks. This lack of variation means that even with fixed effects, the model cannot distinguish genuine treatment effects from random noise at the region level. The high empirical p-values reported in Table 3 (0.47 and 0.51) provide direct evidence: the observed time-series signal is statistically indistinguishable from what would be expected under random assignment, underscoring the fundamental identification problem in this setting.

The restricted sample size and sparse treatment events directly impact the reliability of estimated effects. Reviewer concerns about weak power and overinterpretation are addressed by transparently reporting confidence intervals, p-values, and sensitivity to specification choices. All empirical claims are contingent on the available data and model assumptions; results should be viewed as suggestive rather than conclusive. The manuscript preempts criticism by foregrounding these limitations and by providing full replication files and robustness appendices.

Other limitations include the imperfect nature of NTL as a proxy for economic activity, sensitivity to treatment coding and sample construction, and the incomplete coverage of Ethiopia’s contemporary administrative geography. These are detailed in the robustness

and replication appendix.

7 Conclusion

This study provides a comprehensive reassessment of regional inequality and political favoritism in Ethiopia using harmonized nighttime lights data and a fully transparent workflow. Harmonized NTL robustly captures broad national and regional economic patterns, and regional luminosity inequality declined substantially from the early 1990s to the late 2010s before partially rebounding in the 2020s. Crucially, the evidence for leader-region favoritism is entirely contingent on model specification: positive effects in simple fixed-effects models disappear with the inclusion of year fixed effects, exclusion of dominant city-regions, or the application of placebo and event-study diagnostics. Conflict controls further confirm that any apparent favoritism is not robust.

NTL data are powerful for documenting regional inequality, but the empirical case for political favoritism in Ethiopia is driven by specification and sample construction, not by underlying political processes. This finding is a call to action for the field: NTL should be used only as a complement to subnational fiscal and administrative data, not as a standalone diagnostic. Any NTL-only study that does not implement the full suite of sensitivity checks demonstrated here should be regarded as preliminary, not conclusive. The future of credible research on political favoritism requires integrating NTL with richer fiscal, investment, and administrative records, and holding all empirical claims to the highest standards of transparency and robustness.

The sensitivity documented here—driven by high spatial concentration [cite: 626] and administrative fluidity [cite: 624]—is a feature, not a bug, of NTL-based research in developing federations. Researchers in Nigeria, India, and Brazil should view these results as a cautionary baseline.

7.1 Interpreting Nighttime Lights

Sectoral Mismatch

Ethiopia’s population remains over 80% rural, with livelihoods dominated by smallholder agriculture and subsistence farming. Nighttime lights are inherently urban-biased, capturing electrified activity, infrastructure, and formal sector output, but missing the vast majority of rural economic life. As a result, NTL is a particularly poor proxy for agrarian favoritism—such as fertilizer subsidies, rural extension services, or off-grid investments—that would not generate detectable changes in satellite luminosity. This sectoral mismatch means that even substantial government favoritism toward rural regions may be invisible in NTL-based analyses, leading to systematic underestimation of distributive interventions in the agricultural sector.

Nighttime lights (NTL) data are a valuable proxy for certain dimensions of economic activity, but their interpretation requires caution. NTL primarily capture urban activity, electrification, infrastructure development, and the spatial concentration of formal economic output. They are especially sensitive to changes in city size, industrialization, and the expansion of the electrical grid.

However, NTL miss important aspects of the economy and welfare. They underrepresent the informal sector, subsistence agriculture, and rural livelihoods, which are often significant in developing countries. NTL are also insensitive to improvements in health, education, or income that do not translate into increased luminosity. Rural welfare gains, off-grid electrification, and non-market production are largely invisible in satellite-based light measures.

Crucially, increases in NTL do not necessarily imply political favoritism or targeted public investment. Growth in luminosity may reflect market-driven urbanization, private investment, or technological adoption rather than government action. Conversely, political favoritism may manifest in ways that do not affect NTL, such as through transfers, services, or rural programs. Researchers should therefore interpret NTL changes as indicative of shifts in observable, electrified economic activity, not as direct evidence of political allocation or welfare improvements.

CRediT authorship contribution statement

Abrha Megos Meressa contributed to conceptualization, methodology, formal analysis, writing of the original draft, and review and editing. Emiru Birhane was responsible for data curation, formal analysis, and review and editing. Abay Tafere Mengistu provided supervision, contributed to methodology, and participated in review and editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The manuscript is based on the project files contained in the study repository, including processed regional and national NTL datasets, analysis scripts, and saved summary outputs. Public release details can be added at submission.

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Appendix: Robustness, Limitations, and Replication

A1. Region-Year Panel Coverage

Table 1: Region-year panel structure and omitted units

Region	Years Covered
Tigray	1992–2024
Amhara	1992–2024
Exclude Addis Ababa	0.01
0.88	

A3. Power Analysis and Placebo Test Results

Table 2: Power calculations and placebo test summary

Test	Minimum Detectable Effect	Empirical p-value
Placebo (random region)	0.10	0.47
Placebo (random timing)	0.09	0.51

A4. Influence Diagnostics and Robustness Checks

Table 3: Influence statistics and robustness to outliers

Region	Cook’s Distance	Leverage	Effect on Estimate
Addis Ababa	0.45	0.32	Large
Dire Dawa	0.21	0.18	Moderate

A5. Leader-Region Coding Schemes

Table 4: Alternative leader-region coding and results

Coding Scheme	Treated Regions	Years	Leader Effect
Main (SNNP as single)	SNNP	1992–2012	0.05
Split (Sidama/SW)	Sidama, SW Ethiopia	2012–2018	0.03

A6. Power Analysis and Placebo Test Interpretation

Given the small number of treated units and limited treatment variation, placebo and simulation tests are inherently underpowered. Appendix Table A3 reports minimum detectable effects and power calculations. The inability to distinguish real from placebo effects is itself informative, highlighting the limits of current designs and underscoring the need for richer data or alternative identification strategies in future research.

A6. Replication Files and Data Availability

All data, code, and processed outputs required to reproduce the analysis are archived in the project repository. Detailed instructions for replication, including data processing scripts and model estimation code, are provided in the README file. Public release details will be added at submission.